

E-COMMERCE RECOMMENDATION SYSTEM USING COLLABORATIVE FILTERING

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ABSTRACT

This study implements a collaborative filtering recommendation engine using Singular Value Decomposition (SVD) for a local e-commerce platform, achieving RMSE of 0.94 and increasing click-through rate by 18%.

INTRODUCTION

Online shoppers face information overload as product catalogs expand. Recommendation systems address this by personalizing the shopping experience. Collaborative filtering leverages user-item interaction patterns without requiring product content knowledge. This study develops and evaluates a recommendation engine for a Cavite-based online handicraft marketplace with 1,200 users and 3,500 products.

METHODOLOGY

Singular Value Decomposition (SVD) via the Surprise library was implemented on an implicit feedback matrix of 45,000 user-item interactions. Cross-validation used a 5-fold strategy. Hyperparameter tuning: $n_factors$ in [50, 100, 150], regularization in [0.02, 0.05, 0.1]. Evaluation metrics: Root Mean Squared Error (RMSE), Mean Absolute Error (MAE), Precision@10, and Recall@10. A/B testing was conducted with 50 active users over 2 weeks.

RESULTS

SVD achieved RMSE of 0.94 and MAE of 0.71 on the test set. Precision@10 was 0.43 and Recall@10 was 0.38. The A/B test showed an 18% increase in click-through rate versus the non-personalized control group. Best hyperparameters: $n_factors=100$, regularization=0.05.

DISCUSSION

SVD collaborative filtering provides meaningful recommendation quality for a sparse interaction matrix. The cold-start problem for new users remains unresolved and is the chief limitation. Future work should explore hybrid approaches combining SVD with content-based filtering using product category embeddings.